

P1) $X_1, \dots, X_n \sim \text{Unif}(0, 2\sigma)$

$$f(\vec{x}|\theta) = \frac{1}{\theta^n} \prod_{i=1}^n \mathbb{1}_{[0, 2\theta]}(x_i)$$

$$x_i \leq 2\sigma$$

$$\max\left(\frac{x_i}{2}\right) \leq \sigma$$

$$\sigma \leq x_i$$

$$\sigma \leq \min(x_i)$$

See $x \in \mathcal{X}$ ~~xxx~~

$$= \sigma^{-n} \mathbb{1}_{\left\{ \max\left(\frac{x_i}{2}\right) \leq \sigma \leq \min(x_i) \right\}}$$

$$\max\left(\frac{x_i}{2}\right) \leq \sigma \leq \min(x_i)$$

$$\frac{f(\vec{x}|\sigma)}{f(\vec{y}|\theta)} = \frac{\mathbb{1}_{\left\{ \max(x_i) \leq \sigma \leq \min(x_i) \right\}}}{\mathbb{1}_{\left\{ \max\left(\frac{y_i}{2}\right) \leq \sigma \leq \min(x_i) \right\}}} = 1 \Leftrightarrow \max\left(\frac{x_i}{2}\right) = \max\left(\frac{y_i}{2}\right)$$

$$\min(\hat{x}_i) = \min(y_i)$$

$$T(\vec{x}) = \left(\min(x_i), \max(x_i/2) \right)$$

$$T(X) = (\min(X_i), \max(X_i))$$

e) suficiente minimal

P2) $X_1, X_2 \stackrel{\text{ind}}{\sim} f(x|\alpha) = \alpha x^{\alpha-1} e^{-x^\alpha} \quad x \geq 0, \alpha > 0$

$$T(x_1, x_2) = \frac{\log(x_1)}{\log(x_2)}$$

$$Y_1 = \log(x_1) \text{ ind}$$

$$Y_2 = \log(x_2)$$

$$\begin{aligned} e^{Y_1} &= x_1 \\ e^{Y_1} &= \frac{\partial x_1}{\partial Y_1} \end{aligned}$$

$$f_{Y_1}(y_1) = f_x(h^{-1}(y_1)) \frac{\partial x_1}{\partial y_1} = 2 \exp\{2y_1 - e^{2y_1}\}$$

$$Z = 2Y_1$$

$$\frac{Z}{2} = Y_1$$

$$\frac{1}{2} = \frac{\partial Y_1}{\partial Z}$$

$$f_Z(z) = \exp\{z - e^z\}$$

no depende de 2

$$T(x_1, x_2) = \frac{Y_1}{Y_2} = \frac{Z/2}{Z_2/2} = \frac{Z_1}{Z_2} \rightarrow \text{no depende de } \alpha. \rightarrow T \text{ es ancila}$$

P3) $X_1, \dots, X_n \sim \text{Poisson}(\theta)$

$$T = \sum_i X_i \sim \text{Poisson}(n\theta)$$

P4) Sea h función

$$E(h(T)) = 0 \quad \forall \theta \rightarrow h(\bar{c}) = 0 \quad \forall \bar{c} \in T$$

$$E(h(T)) = \sum_{T=0}^{\infty} h(T) \frac{e^{-n\theta} (n\theta)^T}{T!} = \underbrace{e^{-n\theta}}_{\forall \theta} \sum_{T=0}^{\infty} \frac{h(T) (n\theta)^T}{T!} \rightarrow \sum_{T=0}^{\infty} \frac{h(T) (n\theta)^T}{T!} \xrightarrow{\geq 0} h(T) = 0 \quad \forall T \in [0, \infty)$$

no puede ser que h(T) = 0

P3) $X_1, \dots, X_n \sim f(x|h) = e^{-(x-h)} \quad -\infty < h < x < \infty$

$$f(\vec{x}|h) = e^{-\sum x_i - nh} \prod_{i=1}^n \mathbb{1}_{(h, \infty)}(x_i)$$

$$= e^{-\sum x_i - nh} \mathbb{1}_{\{h < \min(x_i)\}}$$

$h < x_i \quad \forall i$
 $h < \min(x_i)$

Sea $\vec{x} \in \mathcal{X}$

a) $\frac{f(\vec{x}|h)}{f(\vec{y}|h)} = e^{-\sum x_i - \sum y_i} \frac{\mathbb{1}_{\{h < \min(x_i)\}}}{\mathbb{1}_{\{h < \min(y_i)\}}} = K \quad (-) \quad T(\vec{x}) = \min(x_i)$

b) $\Pr(X_{(1)} \leq z) = [1 - \Pr(X_{(1)} \geq z)] = 1 - \prod_{i=1}^n \Pr(X_i \geq z)$

$$= 1 - \hat{\prod}_{i=1}^n [1 - \Pr(X_i \leq z)]$$

~~$\hat{\prod}_{i=1}^n$~~

$$F_X(z) = \int_h^z e^{-(x-h)} dx$$

$$e^h \int_h^z e^{-x} dx$$

$$e^h \cdot (-e^{-x}) \Big|_h^z = e^h (e^{-h} - e^{-z}) = 1 - e^{-(z-h)} = F_X(z)$$

$\frac{\partial}{\partial t} \int_{X_{(1)}} F_{X_{(1)}}(z) = 1 - [1 - 1 + e^{t+h}]^n = 1 - e^{-n(z-h)}$

$$f_{X_{(1)}}(z) = n e^{-n(z-h)} \quad t > h$$

$$E(h(\vec{X})) = \int_h^\infty h(\vec{z}) n e^{-n(z-h)} dz = 0$$

$$n e^{nh} \int_h^\infty h(\vec{z}) e^{-nz} dz = 0$$

$$\frac{d}{dh} \int_h^\infty h(\vec{z}) e^{-nz} dz = 0$$

~~$-h(h) e^{-nh} \geq 0$~~

$-h(h) = 0 \quad \forall h$
 $h(h) = 0 \quad \forall h$

No condici
por lo tanto
siempre es 0
en todo
 $z \in T$.

Por ende $X_{(1)}$ es completo

13)
$$S^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$= \frac{1}{n} \sum_{i=1}^n ((x_i - \mu) - (\bar{x} - \mu))^2$$

$$= \frac{1}{n} \sum_{i=1}^n \left((x_i - \mu) - \left(\frac{1}{n} \sum_{i=1}^n (x_i - \mu) \right) \right)^2$$

$$= \frac{1}{n} \sum_{i=1}^n (z_i - \bar{z})^2$$

$\downarrow \quad \quad \downarrow$
 no depende de μ

$$z_i = x_i - \mu$$

$$z_i + \mu = x_i$$

$$z_i \sim f(z | \mu=0) \rightarrow \text{no depende de } \mu$$

$$\bar{x} = \frac{1}{n} \sum x_i$$

$\rightarrow \int^2$ no depende de μ

luego $\int^2$ es un estadístico ancilar

Luego por "Teorema de Basú" $X(1)$ y $\int^2$ son independientes

14) $X_1, \dots, X_n \sim f(x|\theta) = \sigma x^{\theta-1} \mathbb{I}_{(0,1)}(x)$

a)
$$\frac{f(\vec{x}|\theta)}{f(\vec{x}|\eta)}$$

$$= \frac{\sigma^n (\prod x_i)^{\theta-1}}{(\sigma)^n (\prod x_i)^{\eta-1}} = \frac{\sigma^n (\prod x_i)^{\theta-1}}{(\sigma)^n (\prod x_i)^{\eta-1}} \exp\{\theta \sum \log(x_i) - \eta \sum \log(x_i)\}$$

suficiente minimal completo

$\downarrow \quad \quad \downarrow$
 $(\theta) \quad \eta(\eta) \quad \quad \eta_1(\eta) = 0 \quad T_1(\eta) = \sum \log(x_i) = \log(\prod x_i)$

familia exponencial

b) $Y_1 = \log(x_1)$

$$e^{Y_1} = x_1$$

$$\frac{\partial x_1}{\partial Y_1} = e^{Y_1}$$

$$f_{Y_1}(\eta|\theta) = \sigma e^{Y_1(\theta-1)} \cdot e^{Y_1}$$

$$f_Y(Y_1|\theta) = \sigma e^{Y_1 \theta}$$

$\mathcal{C} = \prod x_i$ También

e) suficiente minimal completo

$z_1 = \theta Y_1$

$$f_{z_1}(z|\theta) = \frac{\sigma}{\theta} e^z = e^z \text{ no depende de } \theta$$

$\frac{z_1}{\theta} = Y_1$

$\frac{1}{\theta} = \frac{\partial Y_1}{\partial z_1}$

luego $\frac{\log(x_1)}{\log(x_2)} = \frac{Y_1}{Y_2} = \frac{1/\theta z_1}{1/\theta z_2} = \frac{z_1}{z_2} \rightarrow \text{no depende de } \theta$

$\frac{Y_1 \log(x_1)}{\log(x_2)}$ es ancilar

c) Por Teorema de Basú son independientes